

Memo to the Emergency Task Force:

Will Humankind Survive the Zombie Apocalypse? - Part II

Introduction: After the submission of your previous memorandum, the Chair of the Emergency Task Force decided to allocate all available resources to the development of a vaccine against the Zombie disease. The manufacturer now needs to optimize their process to manufacture as much vaccine as possible.

Technical Details: For the production of the vaccine, our chemical engineers use a biochemical reactor (volume $V = 25m^3$) connected to three pipes. The first pipe will introduce a solution with the virus, the second a chemical that inactivates the virus and the third pipe will collect the final product. The discharge for the first and second pipes is $Q_1 = Q_2 = 2.5 \frac{m^3}{min}$ and $Q_3 = Q_1 + Q_2 = 5 \frac{m^3}{min}$ for the third one. The virus and chemical concentrations of the solutions in the pipes will be of $c_{z0} = 20 \frac{mg}{m^3}$ and $c_{c0} = 25 \frac{mg}{m^3}$ respectively. Despite the rush and lack of materials, the reactors are build to very high precision and you do not need to concern yourselves with uncertainties in the reactor specifications.

The “inactivation” reaction between the virus and the chemical is as follows:



where Z represents the virus, C the chemical and P the inactivated virus (that is, the vaccine). The reaction rate is constant at $k = 1.2 \frac{m^3}{mg \times min}$.

Task: Your task is to deliver a reliable and efficient model of the reaction which the engineers can use to find optimal operating parameters. To do this, you will have to provide the following deliverables

1. a mathematical model for the reaction described above.¹
2. a critical discussion of the limits of the model.
3. well reasoned suggestions for suitable numerical solution methods considering
 - (a) accuracy,
 - (b) speed,
 - (c) robustness.

Choose the numerical solution method from the following set of candidates:

1. Explicit Euler Method
2. Improved Euler Method
3. Adams-Bashforth-2 (AB2)
4. Adams-Moulton-2 (AM2)
5. Backward Differentiation Formula BDF2

To ensure a fair comparison, you have to implement all these methods by yourself in Python and compare them with respect to accuracy, speed, and robustness as stated above.

You have to give a clear recommendation in your report for one method which you consider best for solving the given problem. Back up your choice by referencing the results you presented as evidence.

¹An inspiration may be found Chapter 28, Section 1, in Chapra, S. C., & Canale, R. P. (2010). *Numerical methods for engineers*. Boston: McGraw-Hill Higher Education (see StudIP Literature). The model will be conceptually similar to the predator-prey model that you already know.

Hints:

1. You can use Python's built-in ODE solvers for estimating global errors, and for generating a second starting value for two-step methods.
2. For implicit methods you must implement a Newton iteration for solving the arising nonlinear systems of equations. Terminate the iteration if the difference of the iterates is appropriately small, eg less than $1e-8$.

Submission: Your submission consists of two parts

1. A technical report of up to 2 pages as PDF submitted to ILIAS, see StudIP for the link.
File name: `LASTNAME_GIVENNAME.pdf`
2. A Python code verified in YAPS.
The YAPS system will be open from **01.06. to 09.06.**
Link: <https://yaps.amh.tuhh.de/numode/project-2-ss26/examserver/client/instance/>

In YAPS you must implement the following five functions:

1. `euler_exp1(f, y0, t)`, Explicit Euler Method
2. `euler_impr(f, y0, t)`, Improved Euler Method
3. `twostep(f, Jf, y0, y1, t, a, b)`, general explicit or implicit two-step method, eg AB2, AM2, BDF2
4. `f(t, c)`, right-hand side of the sought ODE $c' = f(t, c)$ for the biochemical reaction
5. `Jf(t, c)`, Jacobian of f with respect to c (needed for Newton iteration in implicit methods)

For all five functions templates with parameter description and examples are given in the file `exercise.py` which you can download in StudIP/ILIAS. You shall complete these templates.

Before you submit your functions in YAPS you can a priori check them offline with the main program `__main__.py` which is also available in StudIP/ILIAS. (Finally, YAPS will run additional tests of your code which we do not disclose.)

Deadline: 09.06.2026

Dr. wich. tig. N. Otreál
Chair of the Emergency Task Force

Report structure. Please structure your report using the following sections.

1. Introduction: this should briefly describe the problem, introduce the model, explain the question to be answered and describe how the interventions are modeled.
2. Scenarios: present example scenarios that illustrate the possible outcomes you identified.
3. Recommendation: provide your recommendation and reference your scenarios to provide evidence.

Please use 10pt or 11pt fontsize and a standard font like Arial, Helvetica or something similar. You must respect the limit of two pages, anything beyond two pages will not be considered for marking.

Some general tips for writing reports.

- Do not mix answers to different questions. Every question deserves its own answer that is clearly separate from the others. Do not let the reader figure out which question you are addressing in a given part of the text.
- Every figure needs to be referenced and discussed in the text. Each figure should have a clear “message” associated with it which should be clearly spelled out in the text. If you can’t say what the message of a figure is, you either need to think more about it before writing it down or the figure doesn’t really have one - remove it, then.

Resist the urge to pad your report with many figures that do not contribute to the key points its making. It will not improve its quality and just look cluttered.

- Keep sentences short. At least one round of proof-reading should be dedicated to identify overly long sentences and to breaking them down.
- At least one additional round of proof-reading should be used to identify and delete waffle. Say what you want and need to say, not more, not less. Being concise is a virtue.
- Use paragraphs.
- Make sure figures are readable, also when printed in black and white. Whenever possible, use vector formats like PDF for graphs as they scale without loss.